

CS & IT ENGINEERING



Operating System

File System

One Shot



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Recap of Previous Lecture



Topic

Inverted Paging

Topic

Hashed Page table

Topics to be Covered



Topic

File System

Topic

Blocks and File Allocation Methods





Topic : File

A file is a named collection of related information that is recorded on secondary storage.





Topic : File Attributes

1. Name
2. Extension
3. Size
4. Date
5. Author
6. Created, Modified, Accessed
7. Attributes: Read-only, hidden
8. Default Program
9. Security Details



Topic : File Directory

→ folder

Collection of files





Topic : File System

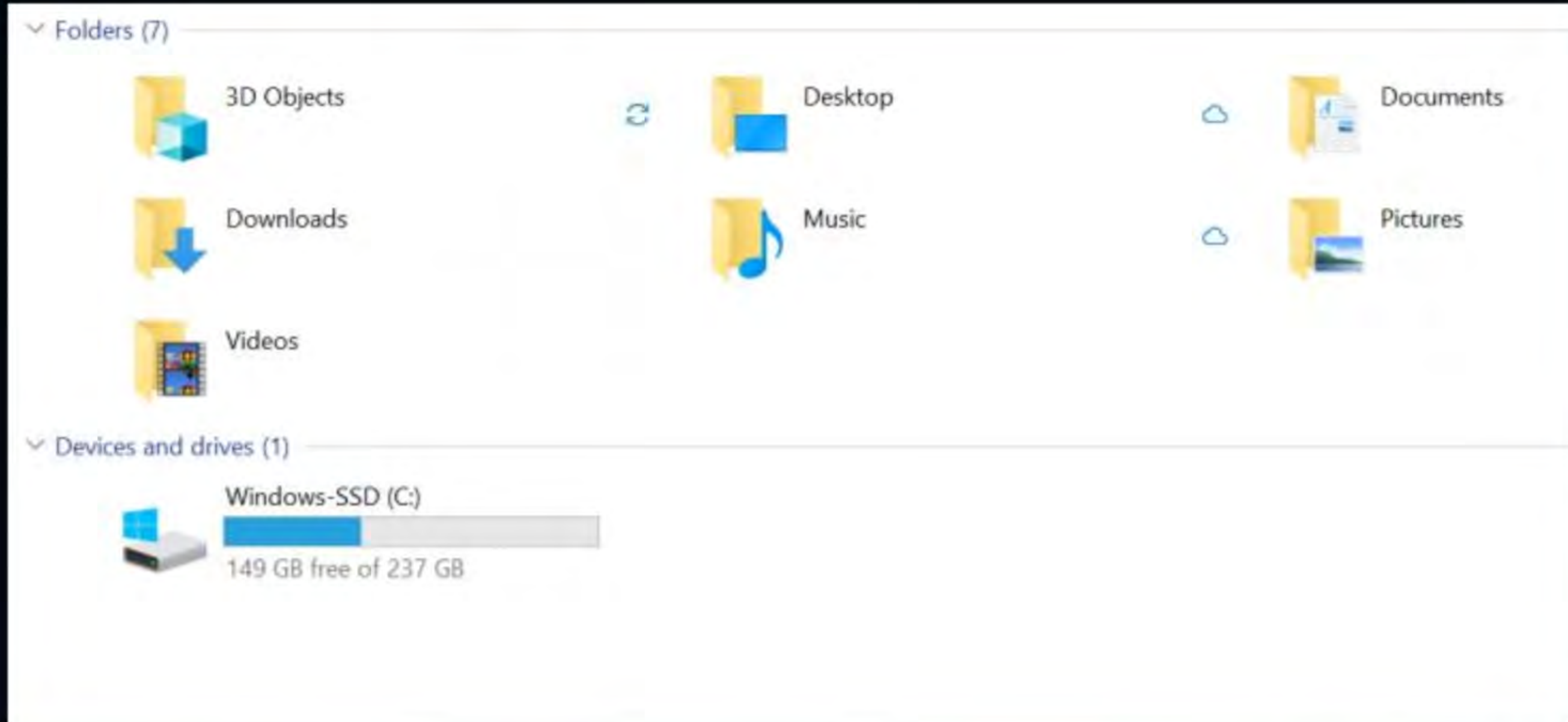
Module of OS which manages, controls and organizes files and related structures

!





Topic : File System





Topic : Types of File Systems

1. FAT32
2. NTFS
3. HFS+
4. Ext2 / Ext3 / Ext4
5. Swap





Topic : File System



Windows-SSD (C:) Properties

Security
General
Previous Versions
Tools
Hardware
Quota
Sharing

Windows-SSD

Type: Local Disk
File system: NTFS

Used space:	93,87,61,70,752 bytes	87.4 GB
Free space:	1,60,84,49,55,648 bytes	149 GB
Capacity:	2,54,72,11,26,400 bytes	237 GB

Drive C: [Disk Cleanup](#)

☐ Compress this drive to save disk space
☒ Allow files on this drive to have contents indexed in addition to file properties

OK Cancel Apply



Topic : Disk Formatting

Physical
(low-level)

making tracks & sectors
on disk

(done by manufacturers)

Logical
(high-level)

→ Creating disk partitions (drives)

→ Install OS in one partition

→ Creating disk block

↳ generally one
block is one sector



Topic : Disk Blocks

↓
smallest logical unit of the disk





Topic : Disk Blocks



Number of disk blocks = 2^{16}

Size of each block = 1KB

Total Size of disk? $= 2^{16} * 1KB$
 $= 2^{26} B$
 $= 64 MB$

→ Disk block no. = 16 bits
(Disk block Address)



Topic : Disk Blocks



Disk block address = 24-bits $\Rightarrow$ no. of blocks = 2^{24}

Size of each block = 2KB

Total Size of disk? = $2^{24} * 2KB$
= $2^{35} B$
= 32GB



Topic : Disk Blocks



Total disk size = 256GB

Block Size = 2KB

Disk block address? = 27 bits

$$\text{no. of blocks in disk} = \frac{256 \text{ GB}}{2 \text{ KB}}$$

$$= 128 \text{ M}$$

$$= 2^{27}$$

↓

$$\text{add.} = 27 \text{ bits}$$



Topic : Free Space Management

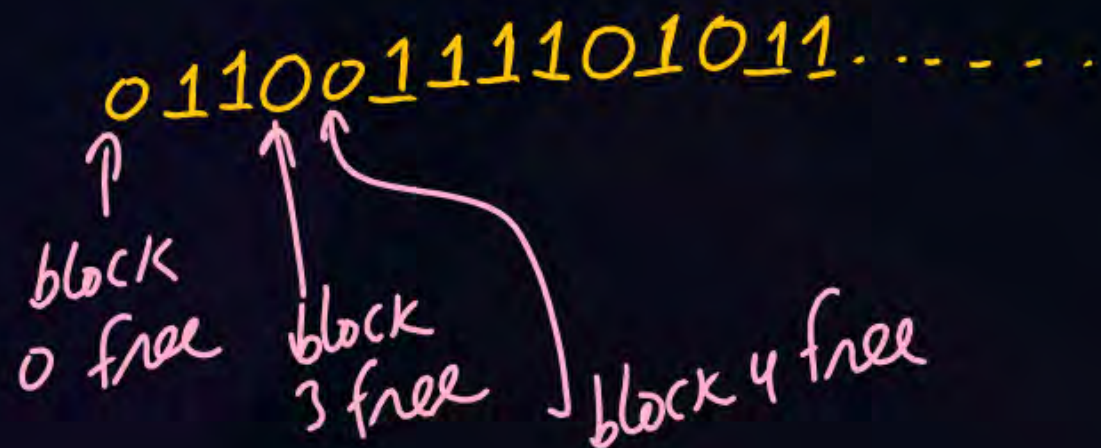
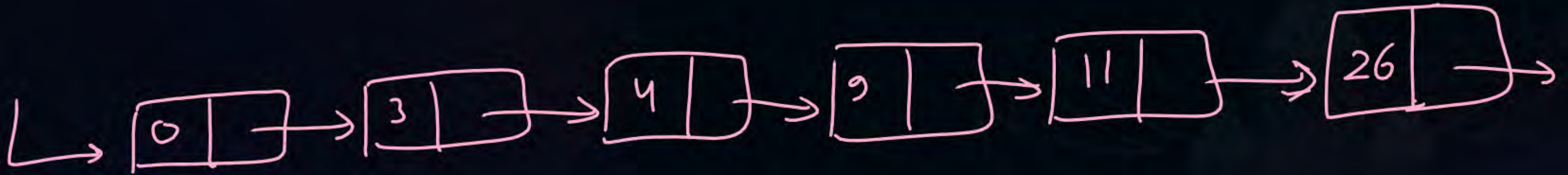
1. Free List

→ linked list containing add. (DBA) of each free block.

2. Bitmap Method

⇒ stores one bit infoⁿ per block

→ 0 free block
→ 1 occupied block





Topic : Free Space Management

1. No searching in free list, but in bitmap we search for first zero *to find a free block*
2. Free list is faster in allocating a free block
3. Free list size is variable, where as bitmap size is constant



Topic : Question

A particular disk unit uses a bit string to record the occupancy or vacancy of its disk blocks with '0' denoting vacant block and '1' denoting occupied block. A 32-bit part of this string has Hexadecimal value of A3D4B055. The percentage of occupied blocks on the disk for this part is ?

$$\begin{array}{r} 1010 \Rightarrow 2 \quad 1^s \\ 2 \\ 3 \\ 1 \\ 3 \\ 0 \\ 2 \\ 2 \\ \hline 15 \end{array}$$

$$= \frac{15}{32} * 100\%$$

$$= 46.875\%$$

Ques) A disk has 2^{24} blocks

$$\text{size of bitmap} = \frac{2^{24}}{8} \text{ bits?}$$

Ques) A disk has 2^{20} blocks each of size $2 \text{ KB.} \rightarrow 2^{11} \text{ B}$

$$\text{no. of blocks needed to store entire bitmap} = \frac{64}{1} \text{?}$$

solⁿ

$$\begin{aligned} \text{size of bitmap} &= 2^{20} \text{ bits} \\ &= 2^{17} \text{ bytes} \end{aligned}$$

$$\begin{aligned} \text{no. of blocks} \\ \text{to store bitmap} &= \frac{2^{17} \text{ B}}{2^{11} \text{ B}} = 2^6 \\ &= 64 \end{aligned}$$



Topic : File Allocation Method



How file is stored on disk on multiple blocks.



Topic : File Allocation Method

1. Contiguous Allocation
2. Linked Allocation
3. Indexed Allocation



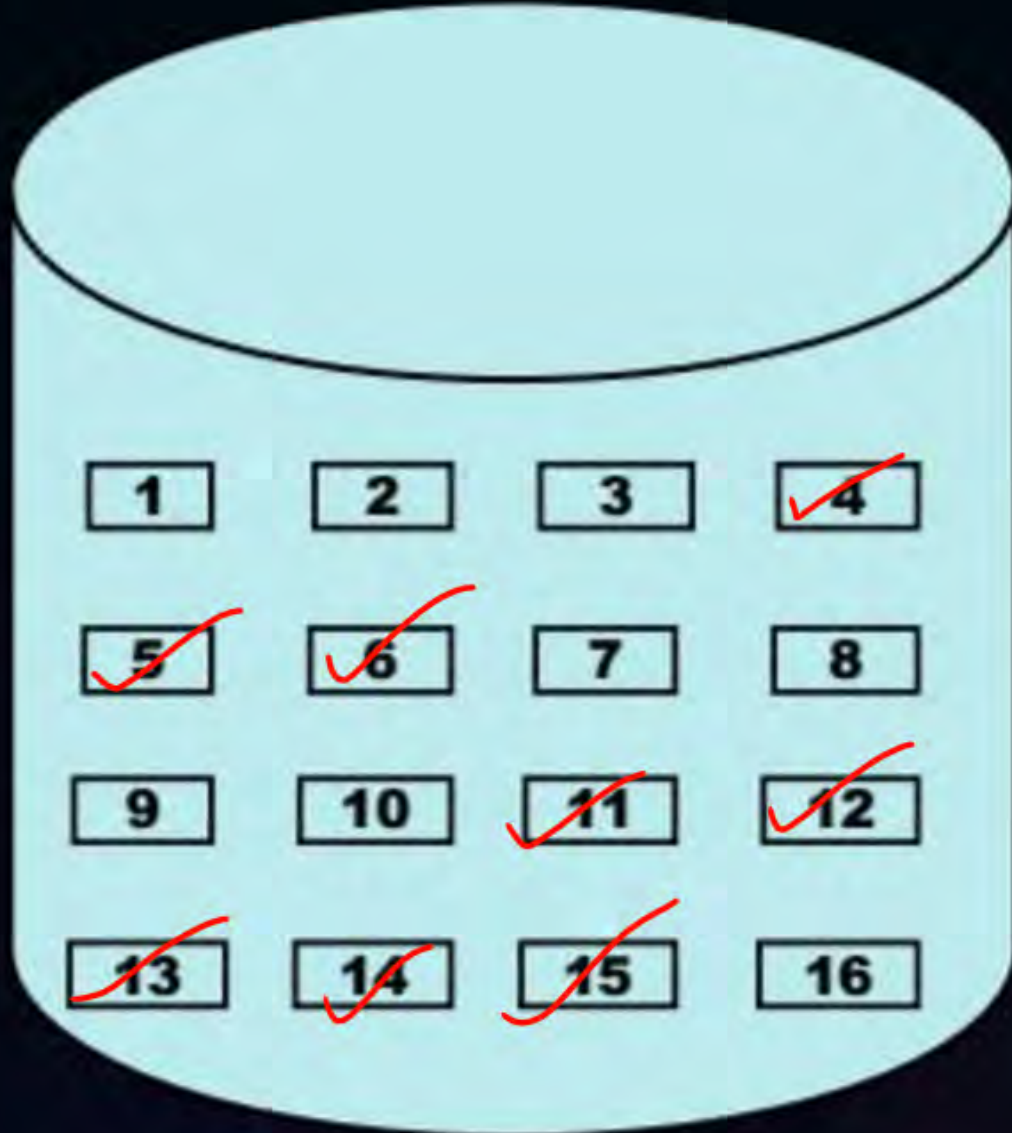


Topic : Contiguous Allocation

→ file is stored on consecutive blocks always.

File Allocation table

File name	starting block	No. of blocks
os.pdf	4	3
COA short notes	11	5





Topic : Contiguous Allocation

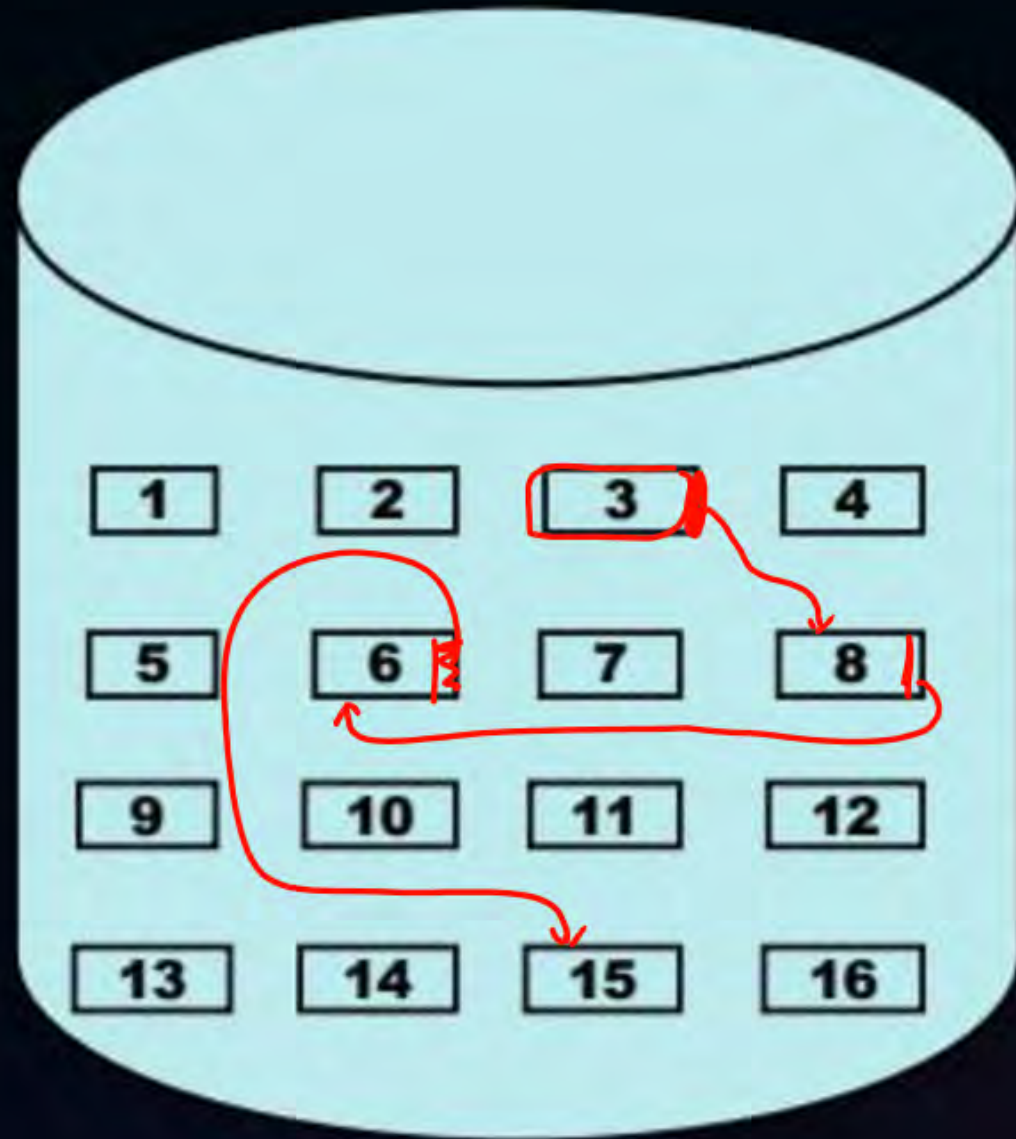
Performance:

1. Fragmentation: Internal, External
2. Increase in File size: Inflexible
3. Type of access: Sequential, Random/direct

Insertion of a new block in file at	<i>No. of blocks accessed</i>
Beginning	<i>Too many</i>
Middle	<i>Too many</i>
End	<i>less</i>



Topic : Linked Allocation



<u>filename</u>	<u>starting block no.</u>	<u>Ending block no.</u>
os.pdf	3	15



Topic : Linked Allocation

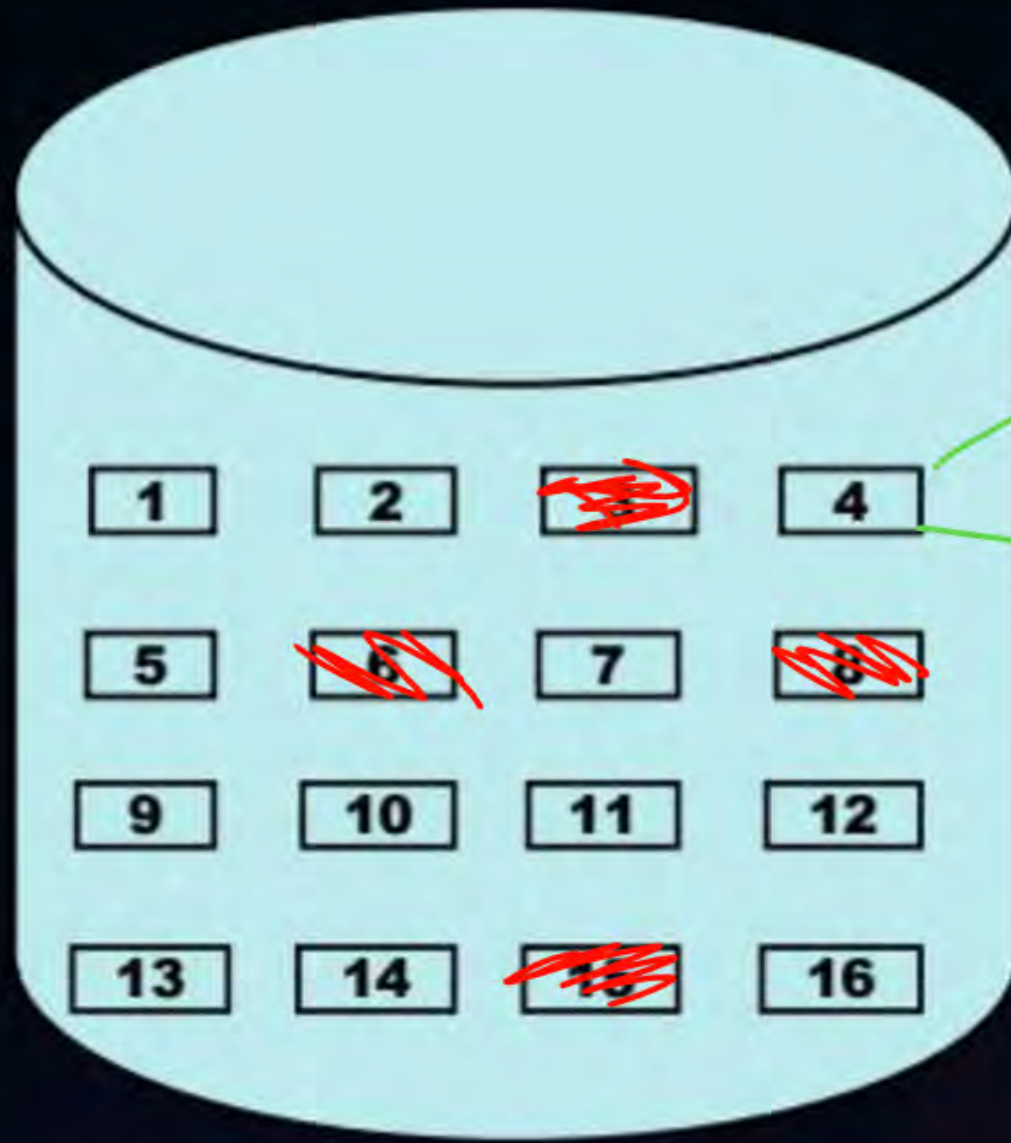
Performance:

1. Fragmentation: Internal
2. Increase in File size: Flexible
3. Type of access: Sequential

Insertion of a new block in file at	<i>no. of block accesses</i>
Beginning	<i>less</i>
Middle	<i>Too many</i>
End	<i>Too many</i>

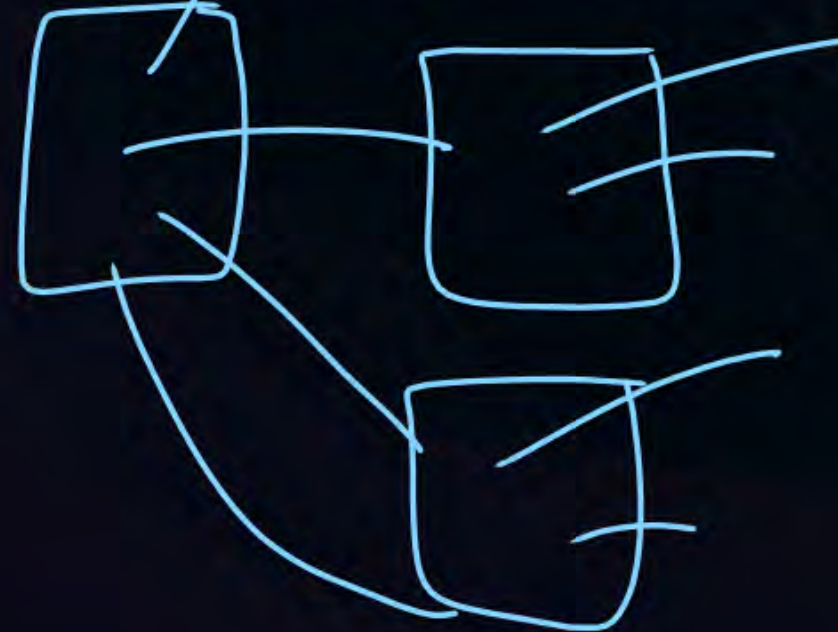


Topic : Indexed Allocation



Index block

3
8
6
15



filename	Index block no.
os.pdf	4



Topic : Indexed Allocation

Performance:

1. Fragmentation: Internal
2. Increase in File size: Flexible
3. Type of access: Sequential, Random/direct

Insertion of a new block in file at	<i>No. of block access</i>
Beginning	<i>Less</i>
Middle	<i>Less</i>
End	<i>Less</i>

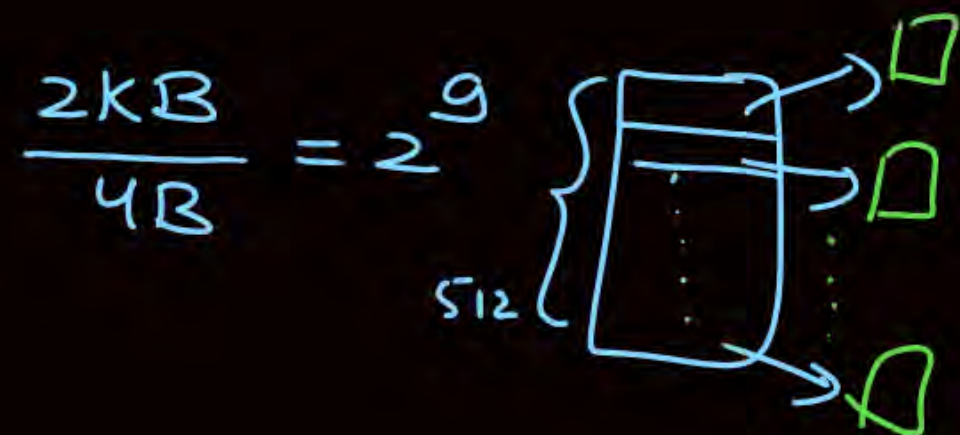
Ques) block size = 2KB

each DBA = 4B

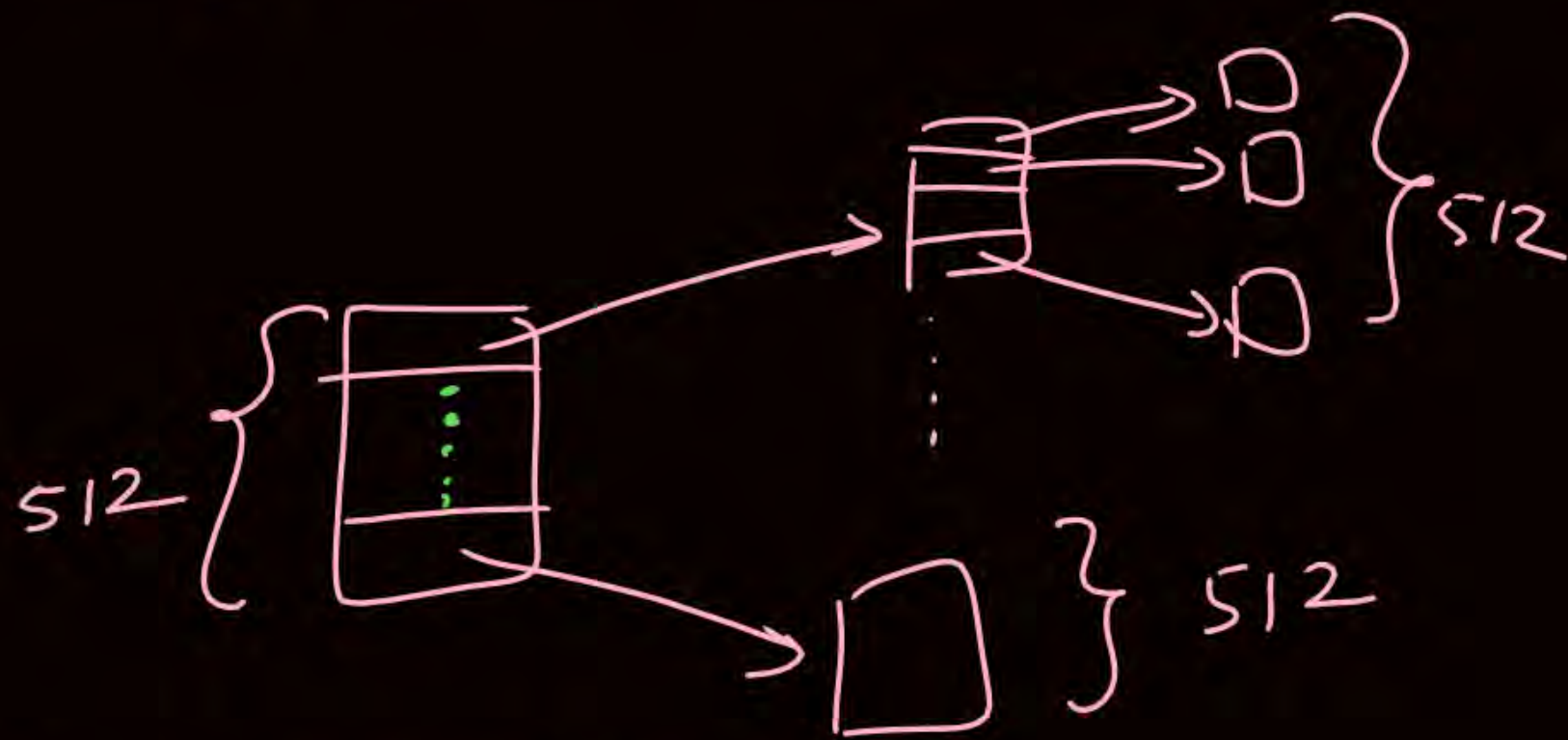
max. no. of indexes per block = $\frac{512}{?}$?

max file size = $\frac{1MB}{?}$?
for single level
indexing.

$$512 * 2KB = 1MB$$



Ques) In prev. Questⁿ,
max file size for 2 level indexing 512 MB?



$$\begin{aligned}\text{max file size} &= 512 * 512 * 2KB \\ &= 2^{29} B = 512 MB\end{aligned}$$



Topic : Question

H.W.

[GATE-2022]



#Q. Consider two files systems A and B , that use contiguous allocation and linked allocation, respectively. A file of size 100 blocks is already stored in A and also in B. Now, consider inserting a new block in the middle of the file (between 50th and 51st block), whose data is already available in the memory. Assume that there are enough free blocks at the end of the file and that the file control blocks are already in memory. Let the number of disk accesses required to insert a block in the middle of the file in A and B are n_A and n_B respectively, then the value of $n_A + n_B$ is_____?



Topic : Question



- Disk block address = 16 bits $\therefore 2^{16}$
- Disk block size = 1KB
- Index block = 1KB
- Maximum file size?

for single-level indexing

$$\text{max indexes in a block} = \frac{1\text{KB}}{2\text{B}} = 512$$

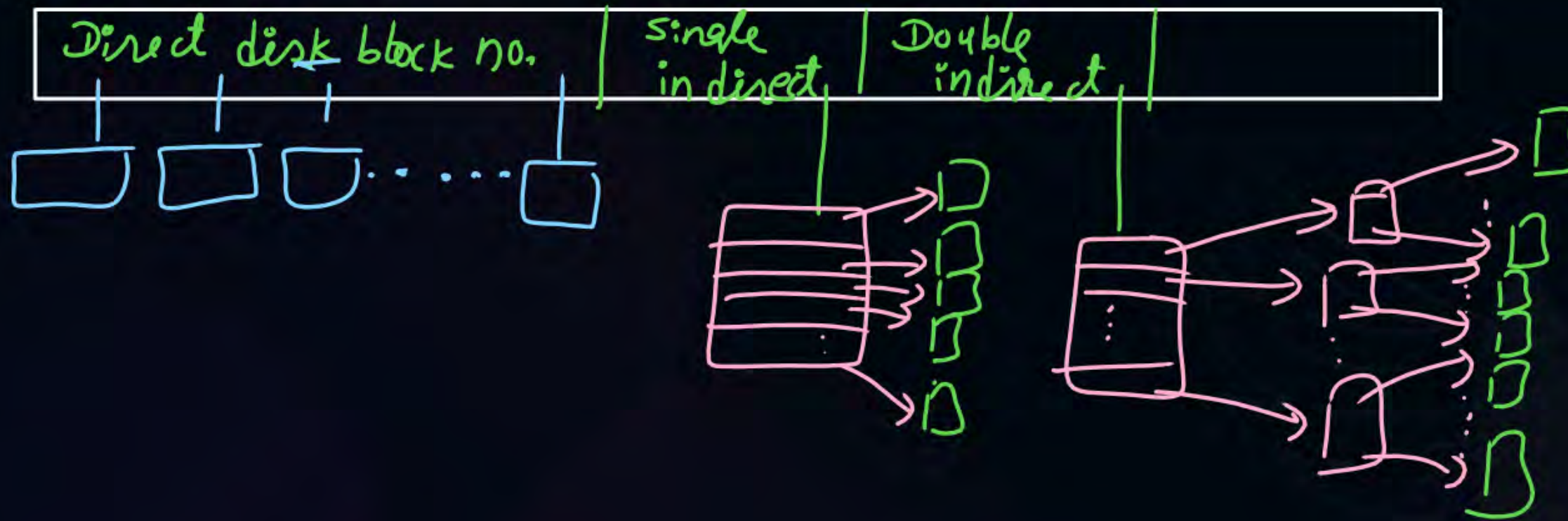
$$\begin{aligned}\text{max file size} &= 512 * 1\text{KB} \\ &= \underline{\underline{512\text{ KB}}} \quad \text{Ans}\end{aligned}$$



Topic : Unix I-node Structure

The inode (index node) is a data structure in a Unix-style file system that describes a file-system object such as a file or a directory.

Indexes in i-node





Topic : Unix I-node Structure

The inode (index node) is a data structure in a Unix-style file system that describes a file-system object such as a file or a directory.

- Each inode stores the attributes and disk block locations of the object's data.
- The number of Inode limits the total number of files/directories that can be stored in the file system.



Topic : Question

[GATE-2019]



#Q. The index node (inode) of a Unix-like file system has 12 direct, one single-indirect and one double-indirect pointer. The disk block size is 4 kB, and the disk block addresses 32-bits long. The maximum possible file size is (rounded off to 1 decimal place) 4.0 GB? $\rightarrow 4B$

$$\text{max indexes per block} = \frac{4KB}{4B} = 1K = 2^{10}$$

$$\text{max file size} = (12 * 4KB) + (2^{10} * 4KB) + (2^{10} * 2^{10} * 4KB)$$

$$= 48KB + 4MB + 4GB$$

$$= 4.004048GB$$



Topic : Question

H.W.

[GATE-2005]



#Q. In a computer system, four files of size 11050 bytes, 4990 bytes, 5170 bytes and 12640 bytes need to be stored. For storing these files on disk, we can use either 100 byte disk blocks or 200 byte disk blocks (but can't mix block sizes). For each block used to store a file, 4 bytes of bookkeeping information also needs to be stored on the disk. Thus, the total space used to store a file is the sum of the space taken to store the file and the space taken to store the book keeping information for the blocks allocated for storing the file. A disk block can store either bookkeeping information for a file or data from a file, but not both.

What is the total space required for storing the files using 100 byte disk blocks and 200 byte disk blocks respectively?

A

35400 and 35800 bytes

B

35800 and 35400 bytes

C

35600 and 35400 bytes

D

35400 and 35600 bytes



Topic : Solution

File Size	No. of blocks to store file	No. of blocks to store bookkeeping	Total blocks	Total Size
11050 Bytes				
4990 Bytes				
5170 Bytes				
12640 Bytes				



2 mins Summary

Topic

File System

Topic

Blocks and File Allocation Methods





Happy Learning

THANK - YOU

